

Executive Product & Architecture Brief

Document Version: 1.0 Focus: Strategy, Data Acquisition & Moat Status: Approved Architecture

1. Executive Overview

The **Government Scheme Guidance Platform (GSGP)** is an intelligent citizen companion built to bridge the last-mile gap in public welfare access. Rather than acting as a static directory, GSGP guides users end-to-end: from pre-application eligibility qualification and 10-language Indic simplification, to risk-prioritized document checklists and post-application AI rejection decoding.

2. Data Acquisition, Sizing & Legal Landscape

Area	Strategy & Institutional Reality
Data Sourcing	• myScheme Portal (myscheme.gov.in): Pre-aggregated database of 1,400+ Central and State schemes under Digital India. • Open Government Data (data.gov.in): Public datasets and ministerial statistics. • API Setu (apisetu.gov.in): National gateway for open government welfare APIs. • Ministerial Guidelines & Gazette (egazette.gov.in): Statutory policy circulars.
Licensing & Costs	₹0 (100% Free) . Under the National Data Sharing & Accessibility Policy (NDSAP) and the RTI framework, public welfare scheme guidelines are public goods. There are zero government royalties or commercial licensing fees.
Dataset Sizing	~250 MB to 500 MB total in structured JSON/text for all ~3,000 Central and State schemes across India. Even with original guideline PDFs archived, total storage is strictly under 10 GB.

3. Technical & Economic Sustainability

To eliminate bill spikes while serving millions of citizens, GSGP operates on a two-tier compute pipeline:

Tier 1: Deterministic SQL Engine (Zero LLM Cost)

Filters 3,000+ schemes down to 3–5 candidate schemes using simple demographic queries (Age, State, Annual Income, Category).

Tier 2: Targeted LLM & Multi-Layer Caching

Invokes LLMs strictly for unstructured tasks (decoding scanned rejection letters, Indic translations). In-memory caching (1–24 hr TTL) eliminates ~85% of redundant calls.

4. Automated Scheme Update Pipeline

Governments continuously revise income brackets and policy criteria. GSGP uses a reliable 3-stage automated pipeline:

[Govt Portals / Ministerial RSS] | (Daily Automated Cron Poller) ▼ [Content Hash & ETag Diff Detection] | (Change Flagged) ▼ [AI Diff Agent] → Compares old vs new text to extract JSON rule revisions | ▼ [1-Click Human Verification] → Moderator clicks "Approve" (guarantees legal accuracy) | ▼ [Database Committed & In-Memory Cache Invalidated]

5. Registration Strategy: In-App Preparation + Official Handoff

Because statutory welfare disbursements mandate Aadhaar OTP authentication, biometric e-KYC, and sovereign Direct Benefit Transfer (DBT) verification, private apps cannot bypass government portals for final legal submission. GSGP uses a **Hybrid Assisted Preparation Model**:

- **In-App Preparation (GSGP)**: Citizens input their profile once. The app generates a 0–100% eligibility score, validates required documents, and prepares an organized, compliant **"Application Kit"** (PDFs compressed under 2MB).
- **Official Portal Hand-off**: Directs the citizen via verified, scam-free direct links to the official department portal with step-by-step guidance.
- **Offline Scheme Support**: For physical submissions, generates a pre-filled, printable official form ready for Tehsil/Panchayat counter delivery.

6. Business Model & User Retention

- **Profile Vault**: Users save verified records once and receive life-cycle notifications for new welfare benefits across education, housing, farming, and pensions.
- **CSC & Cyber Café Partner Operating System**: Village-Level Entrepreneurs (VLEs) and local cyber cafés use GSGP to process citizen applications 5x faster.
- **CSR & Impact Grants**: Social welfare foundations and corporate CSR wings sponsor platforms that verifiably improve welfare uptake.

7. Competitive Moat: What Sets GSGP Apart

Feature Dimension	Existing Portals (e.g. myScheme)	GSGP Platform (Our Advantage)
Rejection Support	❌ None. Citizens are left stranded when applications are rejected with cryptic codes.	☀️ AI Rejection Decoder : Analyzes rejection letters, isolates root causes, and gives clear re-application/appeal steps.
Language Simplicity	⚠️ Literal, dense bureaucratic translations.	☀️ 12th-Grade Indic AI : Conversational guidance across 10 Indian regional languages (Hindi, Tamil, Marathi, Bengali, etc.).
Document Guidance	⚠️ Static list of document titles.	☀️ Risk-Ranked Checklists : Documents ordered by most common rejection risk, with step-by-step acquisition guides and alternates.
Pre-Application Check	❌ None. Applications fail after months of waiting.	☀️ Pre-Eligibility Scoring (0–100%) : Identifies criteria mismatches prior to submission.

Core Platform Value: Existing government portals tell citizens *what schemes exist*, but abandon them when paperwork is confusing or applications get rejected. GSGP is an end-to-end AI companion that pre-qualifies, simplifies, guides, and decodes rejections so every eligible citizen receives their rightful benefits.